

LEAN-BASED DPP READINESS ASSESSMENT FRAMEWORK

for Small and Medium Bangladeshi Garment (RMG) Factories

A Practical Project Framework Applying Lean Management Tools to EU Digital Product Passport (DPP)
Readiness

Prepared as a Practical Lean Management Capstone Project

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Status: Desk-based / illustrative project using publicly available Bangladesh RMG sector data. No factory-specific data was collected for this version — see Section 4 (Scope & Limitations).

1. Executive Summary

The European Union's Ecodesign for Sustainable Products Regulation (ESPR) — which entered into force on 18 July 2024 — introduces the Digital Product Passport (DPP), a structured digital record of a product's materials, supply chain, environmental footprint, and durability. Textiles and apparel have been identified as a priority sector, with a textile-specific delegated act expected around 2027 and mandatory compliance anticipated from 2027–2028, subject to extended timelines for small and medium-sized enterprises (SMEs).

Bangladesh's Ready-Made Garment (RMG) sector — the world's second largest after China, comprising roughly 4,000 BGMEA-member factories, the majority of which are small or medium-sized — exports heavily to the EU. DPP requirements will therefore directly affect Bangladeshi SME factories that supply EU-facing brands, even though enforcement is still being finalised.

This project proposes a Lean-Based DPP Readiness Assessment Framework: a practical, low-cost diagnostic tool that SME garment factories can use to assess how 'DPP-ready' their internal processes, data systems, and supply chain documentation currently are — and to identify, using Lean tools, the wastes and gaps that stand between their current state and DPP compliance.

The framework combines five Lean tools from the Lean Management curriculum — 5S, Value Stream Mapping (VSM), the 5 Whys, Kaizen, and Kanban — with a DPP Readiness Scorecard covering six readiness dimensions (traceability, material data, supplier data management, chemical/compliance documentation, digital infrastructure, and worker/process documentation culture).

This document is a desk-based academic framework, illustrated with a hypothetical SME case ('Green Stitch Apparels Ltd.') built from publicly available Bangladesh RMG sector statistics. It is designed so a student can either (a) present it as a conceptual framework, or (b) apply the included scorecard and worksheets to a real factory if access becomes available later.

2. Background & Rationale

2.1 What is the EU Digital Product Passport (DPP)?

The Digital Product Passport is a digital record — typically accessed via a QR code, RFID tag, or similar — that travels with a product and contains structured information about its materials, origin, manufacturing, environmental impact, reparability, and end-of-life handling. It is a core instrument of the EU's Ecodesign for Sustainable Products Regulation (ESPR), which entered into force on 18 July 2024.

2.2 Why this matters now (verified timeline)

Based on currently published sources (as of mid-2026), the indicative DPP timeline for textiles is:

- 2024 — ESPR entered into force (18 July 2024), establishing the legal framework for the DPP across priority product groups.
- 2025 — The EU Ecodesign & Energy Labelling Working Plan 2025-2030 (adopted 16 April 2025) confirmed textiles/apparel as a priority category, with an indicative delegated act adoption around 2027.
- 19 July 2026 — Two concrete near-term milestones: the EU Central DPP Registry is expected to go live, and the ESPR's ban on destroying unsold textile goods takes effect for large companies (micro/small businesses exempt; medium-sized companies covered from 2030).
- 2027 (expected) — The textile-specific delegated act defining exact DPP data requirements is anticipated, followed by an approximate 18-month transition period.

- 2027-2028 — Mandatory DPP compliance for textiles is widely expected to begin in this window, with SMEs generally receiving longer compliance deadlines and additional technical support under ESPR provisions.

Important honesty note: as of June 2026, the exact textile delegated act has NOT yet been published, and timelines have already slipped before. This framework therefore treats 'DPP readiness' as a forward-looking preparedness exercise (data, traceability, documentation habits) rather than a claim that any specific legal deadline is imminent.

2.3 Why Bangladeshi SME garment factories specifically?

Bangladesh is the world's second-largest garment exporter after China, with the sector employing approximately 4 million workers and earning roughly \$40 billion annually, mainly from exports to the United States and Europe. The Bangladesh Garment Manufacturers and Exporters Association (BGMEA) has around 4,000 member factories, of which a large share are small or medium-sized enterprises (SMEs) — many with fewer than 1,000 workers and limited digital infrastructure.

Large, well-resourced factories (including Bangladesh's 200+ LEED-certified 'green factories') are generally better positioned to absorb new EU compliance requirements. SMEs, however, often lack: digital record-keeping systems, structured supplier documentation, standardised material data, and dedicated compliance staff. A practical, low-cost, Lean-based self-assessment tool can help these factories begin closing that gap before DPP requirements become mandatory.

3. Project Objectives

1. To map the data and documentation requirements implied by the EU Digital Product Passport (DPP) onto practical, factory-floor readiness dimensions relevant to SME RMG factories.
2. To apply five core Lean management tools (5S, VSM, 5 Whys, Kaizen, Kanban) as diagnostic and improvement instruments for closing DPP readiness gaps.
3. To design a simple, reusable DPP Readiness Scorecard that an SME factory manager (or a student researcher) can complete in under a day, without specialised software.
4. To illustrate the framework's application through a hypothetical but realistic SME case, built on publicly verifiable Bangladesh RMG sector data.
5. To produce a 30/60/90-day Kaizen improvement roadmap template that a factory could realistically follow to improve its DPP readiness score.

4. Scope & Limitations

4.1 Scope

- This is a desk-based academic framework developed for a Lean Management practical project (SEU, 3rd year).
- It focuses on apparel/textile SME factories in Bangladesh exporting (directly or via buying houses) to EU markets.
- It covers internal process readiness only — not legal certification, which can only be granted once the EU textile delegated act and Bangladesh-side implementation guidance are published.

4.2 Limitations (stated honestly)

- No real factory was visited or audited for this version of the project. All factory-level figures used in Section 8 (Illustrative Case) are hypothetical and clearly labelled as such.

- EU sector-level statistics, BGMEA figures, and ESPR/DPP timelines cited in this document are drawn from publicly available sources referenced in Section 11 and reflect information available as of mid-2026; exact compliance dates may change as delegated acts are finalised.
- The DPP Readiness Scorecard (Section 6) is an original synthesis created for this project based on publicly described DPP data categories (material composition, traceability, chemical compliance, durability/repairability, supply chain data). It is NOT an official EU or BGMEA compliance checklist, and should be presented as such.
- If the student gains access to a real factory later, Sections 6-9 can be re-run with real data with minimal changes to the framework.

5. Methodology — Mapping Lean Tools to DPP Readiness

Each Lean tool from the Alison Lean Management course is mapped to a specific role in assessing or improving DPP readiness, as summarised below.

Lean Tool	Traditional Lean Use	Applied to DPP Readiness
5S	Sort, Set in order, Shine, Standardise, Sustain — workplace organisation	Applied to DATA & DOCUMENTS instead of physical tools: sorting which material/supplier records exist, where they are stored, standardising naming/formats, and sustaining update habits — the foundation for any DPP data feed.
Value Stream Mapping (VSM)	Maps material and information flow from supplier to customer to find waste	Maps the CURRENT information flow for one product line — from yarn/fabric supplier through cutting, sewing, finishing, packing, to export — overlaying where DPP-relevant data (origin, composition, certifications) is or isn't captured at each step.
5 Whys	Root-cause analysis of a defect or problem	Used on each readiness GAP identified in the scorecard (e.g., 'fabric composition is not recorded per batch') to drill down to the real root cause (e.g., supplier invoices don't separate fibre blends).
Kaizen	Continuous, incremental improvement involving all staff	Converts scorecard gaps + 5 Whys findings into a 30/60/90-day improvement roadmap with small, low-cost steps (templates, checklists, simple spreadsheets) rather than expensive software purchases.
Kanban	Visual pull-system for managing work-in-progress	Used to manage the IMPLEMENTATION of the readiness roadmap itself — a simple visual board (To Do / In Progress / Done) tracking each documentation/data improvement task across departments.

6. DPP Readiness Scorecard (Original Tool)

This scorecard was created for this project by synthesising publicly described DPP data categories (material composition, traceability/origin, chemical compliance, durability, and supply chain transparency) into six factory-level readiness dimensions. It is intended as a simple self-assessment, scored 1 (not ready) to 5 (fully ready) per dimension.

Readiness Dimension	Key Assessment Question	Scoring Guide (1-5)
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Traceability & Origin Data	Can the factory trace a finished garment back to fabric lot, dye lot, and supplier?	1 = no records / 5 = batch-level digital records linked to POs
Material Composition Data	Is fibre composition (%cotton, %polyester, etc.) recorded per fabric roll/order?	1 = only on care labels / 5 = recorded digitally per order with supplier certificates
Supplier Data Management	Are supplier details, certifications (e.g., OCS, GRS, GOTS) and contact data centrally stored?	1 = paper files only / 5 = centralised digital supplier database, updated regularly
Chemical & Compliance Documentation	Are chemical inputs (dyes, finishes) and compliance certificates (REACH, OEKO-TEX) logged per order?	1 = not tracked / 5 = logged per order with expiry tracking
Digital Infrastructure & Data Capture	Does the factory use any digital system (ERP, spreadsheets) for production/order data, or is it fully paper-based?	1 = fully paper / 5 = integrated digital system across departments
Documentation Culture & Staff Awareness	Do floor supervisors and merchandisers understand WHY data accuracy matters (beyond audits)?	1 = no awareness / 5 = trained staff, clear SOPs, regular review

Total possible score: 30 (6 dimensions x max 5). Suggested interpretation bands:

- 6-12: Early stage — major Lean/5S foundation work needed before any DPP pilot is feasible.
- 13-21: Developing — core data exists but is fragmented, paper-based, or inconsistent.
- 22-30: Approaching readiness — digital records largely exist; focus shifts to standardisation and supplier integration.

7. Step-by-Step Application Guide (2-Hour Plan)

This is the practical execution plan you can follow today (10 June 2026) to complete a presentable version of this project.

Step 1 (15 min) — Confirm scope & choose your case

- Decide: hypothetical case (use the 'Green Stitch Apparels Ltd.' example in Section 8 as-is), OR substitute a factory you know informally (even from news/secondary sources) — but do NOT invent factory-specific numbers and present them as real.
- State your scope clearly on a cover slide/page: 'Desk-based Lean framework, illustrated with a hypothetical SME case.'

Step 2 (20 min) — Read & adapt Sections 2-4

- Read the Background section so you can explain the EU DPP timeline in your own words during a viva/presentation.
- Add your own name, student ID, course code, and supervisor name to the cover page.

Step 3 (30 min) — Walk through the Lean tool mapping (Section 5) and DPP Scorecard (Section 6)

- For each of the 6 scorecard dimensions, assign a score (1-5) for the hypothetical case OR your chosen real-world reference factory based on typical SME conditions described in industry reports.

- Write 1-2 sentences justifying each score — this becomes your 'findings' section.

Step 4 (25 min) — Run a mini VSM and one 5 Whys on the lowest-scoring dimension

- Sketch a simple current-state information flow (5-7 boxes: Fabric Supplier -> Cutting -> Sewing -> QC -> Finishing -> Packing -> Export) and mark where DPP-relevant data is/isn't captured.
- Pick the lowest-scoring dimension and run a 5 Whys (template provided in Section 8.4) to find its root cause.

Step 5 (20 min) — Build the Kaizen roadmap & Kanban board

- Use the 30/60/90-day template (Section 9) and fill in 2-3 realistic actions per period.
- Sketch a simple 3-column Kanban (To Do / In Progress / Done) listing the same actions — a hand-drawn photo or simple table is sufficient.

Step 6 (10 min) — Finalise & export

- Review Section 10 (Recommendations) and Section 11 (References) — add any extra sources you used.
- Export to PDF (Word: File > Save As > PDF, or Google Docs: File > Download > PDF) and submit.

Total: approximately 2 hours. If you have more time, the highest-value addition is replacing the hypothetical scorecard scores in Section 8 with scores from a real conversation with anyone working in an RMG factory (even informally) — this instantly upgrades the project from 'conceptual' to 'field-informed.'

8. Illustrative Case Application — “Green Stitch Apparels Ltd.” (Hypothetical SME)

IMPORTANT: “Green Stitch Apparels Ltd.” is a FICTIONAL, illustrative SME factory created for this project. It is NOT a real company. Its characteristics (size, location, products) are constructed to be representative of a typical Bangladeshi SME woven/knitwear exporter, based on publicly described sector norms (BGMEA membership profile: ~4,000 factories, ~60% woven / ~40% knit/sweater; many SMEs employ a few hundred to under 1,000 workers). Any resemblance to a real factory is coincidental.

8.1 Hypothetical Factory Profile

- Location: Gazipur Export Processing Zone area (a common cluster for SME knitwear/woven units)
- Size: ~450 workers, one knitwear product line (basic T-shirts and polo shirts) for two EU-facing buying houses
- Current systems: Paper-based cutting/sewing tickets, Excel sheets for order tracking, no centralised ERP
- Export markets: Primarily EU (via buying houses) and UK

8.2 Hypothetical Scorecard Results

Dimension	Score (1-5)	Justification (illustrative)
Traceability & Origin Data	2	Cutting tickets reference fabric rolls, but rolls aren't linked digitally to specific supplier shipments.

Material Composition Data	2	Composition appears on care labels and fabric inspection reports, but isn't logged per order in a central file.
Supplier Data Management	3	A merchandising team keeps a supplier contact spreadsheet, but certifications (OCS/GRS) are stored as scanned PDFs in shared folders, not indexed.
Chemical & Compliance Documentation	3	OEKO-TEX/REACH certificates collected at supplier onboarding but not re-checked for expiry against active orders.
Digital Infrastructure & Data Capture	2	Excel-based order tracking exists; production floor data (cutting/sewing/finishing) is still paper-based.
Documentation Culture & Staff Awareness	2	Staff understand compliance audits (social/safety) well, but are largely unaware of emerging EU data-traceability requirements.

Hypothetical total score: 14 / 30 -> “Developing” band. Core data exists in fragments but is not centralised, standardised, or order-linked — typical of many SME factories per the project's scope assumptions.

8.3 Mini Value Stream Map (Current State — Information Flow)

A simplified current-state information flow for one knitwear order, showing where DPP-relevant data is captured (✓) or missing (✗):

Fabric Supplier	Receiving / Inspection	Cutting	Sewing	Finishing & QC	Packing & Export
Composition + GRS cert sent as PDF (✓ exists, but not linked digitally)	Fabric inspection report logged on paper (✗ not digitised)	Cutting ticket references fabric roll # (✓ partial)	No data capture — sewing line tracks output qty only (✗)	QC checks quality, not composition/origin (✗)	Packing list + commercial invoice (✓ has order-level data only)

Observation: The fabric-level data needed for a DPP (composition, certification, supplier) EXISTS at the start of the chain but is effectively 'lost' between Receiving and Packing because nothing links the fabric roll record to the final shipped order. This is the core gap the 5 Whys below investigates.

8.4 5 Whys — Root Cause of the Traceability Gap

Problem	Fabric composition and certification data is not linked to the final shipped order (lowest-scoring item, Section 8.2).
Why 1?	Because the cutting ticket records the fabric roll number, but this number is not carried forward onto sewing or packing documents.
Why 2?	Because sewing and finishing documents were designed only to track production quantity/output, not material origin.
Why 3?	Because these documents were designed years ago for productivity (Lean takt-time) tracking, before traceability was a buyer requirement.
Why 4?	Because no department 'owns' cross-stage data continuity — each department optimises its own paperwork independently (a classic siloed-process waste).
Why 5 (root cause)?	Because there is no standardised 'order data card' (digital or paper) that travels with the order through every stage — i.e., a 5S 'standardise' gap at the document level.

This root cause is encouraging: it does not require new machinery or large IT investment — it requires a Lean 5S-style standardised document (a simple 'Order Traceability Card', paper or spreadsheet-based) that travels with each order from receiving to packing.

9. Kaizen Improvement Roadmap (30/60/90-Day Template)

This roadmap addresses the root cause identified in Section 8.4 and the other low-scoring dimensions from Section 8.2. Fill in real dates/owners if applying to an actual factory.

Period	Action (low-cost, Lean-style)	Lean Tool Used	Owner / Dept (template)
Days 1-30	Design and pilot a one-page 'Order Traceability Card' (fabric roll #, composition %, supplier, certificate ref) that travels with each cutting batch.	5S (standardise) + Kaizen	Merchandising + Cutting section
Days 1-30	Create a single shared folder structure for supplier certificates (OCS/GRS/OEKO-TEX), named by supplier + expiry date.	5S (sort & set in order)	Compliance / Merchandising
Days 31-60	Train sewing-line and finishing supervisors to record the Order Traceability Card # on existing production sheets (no new system needed).	Kaizen (staff training)	Production Manager
Days 31-60	Re-run the DPP Readiness Scorecard (Section 6) for a follow-up score and compare against the baseline.	PDCA cycle	Project owner / QA
Days 61-90	Pilot a simple shared spreadsheet (Google Sheets/Excel Online) linking Order # → Fabric Roll # → Composition → Certificate ref → Shipment #.	Digital 5S + Kanban tracking	Merchandising + IT/Admin
Days 61-90	Review results with management; decide whether to extend the pilot to a second product line ahead of 2027 EU delegated act publication.	Kaizen review	Management team

9.1 Kanban Board Template (Implementation Tracking)

Use a simple 3-column board (physical whiteboard, sticky notes, or a free tool such as Trello) to track the actions above as they move through implementation:

TO DO	IN PROGRESS	DONE
- Pilot shared spreadsheet (Days 61-90) - Extend to 2nd product line	- Design Order Traceability Card - Train sewing/finishing supervisors	- Create shared certificate folder structure

10. Recommendations & Conclusion

10.1 Recommendations for SME Factories

- Start with documentation, not software: the biggest DPP-readiness gains for SMEs come from standardising existing paper/Excel records (5S), not from buying expensive ERP/traceability platforms.
- Treat DPP readiness as a Kaizen journey: small, continuous improvements (one traceability card, one shared folder) compound over the 18-24 months before mandatory compliance is expected (2027-2028 per current EU timelines).
- Engage buying houses early: many EU brands are already requesting supply-chain transparency data ahead of legal deadlines (commercial pressure often arrives before regulatory enforcement).
- Leverage BGMEA / BUFT / NITER resources: as the national trade body and affiliated training institutions, these organisations are likely channels for future DPP guidance specific to Bangladesh.

10.2 Recommendations for This Project (Next Steps)

- If possible, validate the hypothetical scorecard (Section 8.2) with even one informal conversation with someone working in merchandising, QA, or compliance at any RMG factory — this single step would meaningfully strengthen the project without requiring formal factory access.
- If presenting academically, be transparent (as this document is) about which parts are based on verified public sources (Sections 2 and parts of 4) versus original synthesis created for this project (Sections 6, 8, 9).
- Future versions could pilot the 'Order Traceability Card' concept (Section 9) as a one-page physical/digital template — this alone could be a strong standalone deliverable.

10.3 Conclusion

This framework demonstrates that core Lean management tools — 5S, VSM, 5 Whys, Kaizen, and Kanban — taught in introductory courses such as the Alison Lean Management certification, can be directly and practically applied to a real, emerging 2026-2028 challenge facing Bangladesh's garment industry: EU Digital Product Passport readiness. By reframing 'data and documentation' as the object of 5S and VSM (rather than only physical tools and workspaces), even an SME factory with no digital infrastructure can begin a low-cost, structured journey toward DPP readiness.

11. References

All sources below were accessed in June 2026 and reflect publicly available information at that time. Compliance timelines for the EU DPP are subject to change pending final delegated acts.

- Worldfavor (2025-2026). "EU Digital Product Passport: What it means for the textile industry." blog.worldfavor.com
- Segura (2026). "The EU Digital Product Passport (DPP): All you need to know." segura.co.uk
- TracexTech (2026). "EU Textile Strategy: Digital Product Passport Guide (2026-2027)" and "DPP Product Scope – ESPR Timelines & Requirements." tracextech.com
- Carbonfact (2026). "Digital Product Passport for Textiles: What Fashion Brands Need to Know." carbonfact.com
- Retraced (2026). "How Can You Prepare Your Fashion Brand for the Digital Product Passport (DPP)?" retraced.com

- PassportCraft (2026). “DPP Timeline 2026-2030: Every Product, Every Deadline.” passportcraft.com
- Trimco Group. “What you need to know about the Digital Product Passport.” trimco-group.com
- Portugal Clothing Factory (2026). “Digital Product Passport (DPP): A Practical Guide for Fashion Brands in 2026.” portugalclothingfactory.com
- New Age BD (2025). “BGMEA enrolls 128 new RMG factories.” newagebd.net
- Prothom Alo (2025). “128 new garment factories in 15 months.” en.prothomalo.com
- Sidley Austin LLP / Trade for Development Initiative. “Bangladesh Garment Manufacturers and Exporters Association (BGMEA).” sidley.com
- Wikipedia (2025-2026). “Bangladesh Garment Manufacturers and Exporters Association” and “2025 Dhaka garment factory fire.” en.wikipedia.org
- BGMEA (2021). “A success story: The Bangladeshi garment sector has made remarkable progress in recent years.” bgmea.com.bd
- The Daily Star (2025). “24 garment factories get green certificates in six months: BGMEA.” thedailystar.net
- Daily Industry BD (2025). “Bangladesh sets record with 243 green RMG factories.” dailyindustrybd.com